some agricultural universities to help with developing and deploying drone technology.

The drone technology is targeted to cover a maximum scan distance of 20km in one hour, and provide insights to guide farmers with better sowing, harvesting and nutrient distribution to crops. It is planned to cover one *taluk* (an administrative division of a region) a day, among the 355 *taluks* across 36 districts in the state. The pilot launch is planned in October and November (*rabi* crop season), starting with Latur and Yavatmal districts.

This is not the first time the government has shown an interest in drones for farming. The Indian Council of Agricultural Research (ICAR), in association with other institutes, has been researching drone technologies since 2016, to produce indigenous solutions fit for Indian farmlands, as a part of SENSAGRI project.

Private players are also continuing with proof-of-concept (PoC) projects to show the practical benefits of drone-based farming. Cyient led a PoC of plant nutrient and health analysis on a 40,468sqm (10-acre) land in Mahabubnagar, Telengana, to help farmers optimise the use of fertilisers and irrigation. The PoC led to a 20 per cent improvement in crop yield in a matter of weeks. It has been under expansion across other districts in Telengana.

The limitations

Drones are still under development to ensure quality, durability and performance on the field. Hardware and software are being optimised to ensure data and operational accuracy. Since drones are a relatively new technology for agricultural applications, upfront costs are high right now, starting at around US$ 500 (more than ₹ 35,000), going up to US$ 25,000 (around ₹ 1.77 million). However, Indian technologists, along with aid from the government, are trying to bring down these costs substantially to make drones affordable for farmers, and the projections look promising.

Some companies provide rental business models to help farmers utilise the technology. For instance, Indian startup vDrone Agro provides technical assistance by deploying custom drones across farms as per requirement, scans the land and provides actionable data in all aspects, helping farm owners optimise their yield. It charges on per acre land basis, which can cost between ₹ 400 and ₹ 800 for each acre, variable as per needs.

Connectivity is also a challenge, as wireless or cellular networks in rural areas are still improving. Since drones require a strong wireless connectivity channel, administrations need to focus on rural areas.

India has set a number of stringent federal laws regarding unmanned aerial vehicle (UAV) usage. Rules need to be fine-tuned to ensure easy operations in farms, while maintaining security and ethical usage. Users also have to be made aware of the regulations and their importance.

Finally, spreading awareness among the masses, especially in rural areas, is a major agenda for driving the adoption of such technologies.

2

GREEN TECH:

Shaping A RENEWABLE TOMORROW The DIGITAL WAY

Mankind has found a way to use digital technologies to optimise and improve the usage of renewable energy, by providing the necessary visibility of consumption of fuels to daily users and help them plan better

"I would put my money on the sun and solar energy. What a source of power! I hope we do not have to wait until oil and coal run out before we tackle that," said Thomas Edison, according to a memoir by James D. Newton. The statement seems simple at first, especially remembering the fact that it was said in the first half of the 20th century. But almost a century later, it proves to be a cautionary sign hung in the office of every aware enterprise and nation leader across the world.

We are all talking about renewable energy today, focussing on spreading awareness about its benefits to the masses as an alternative to exhaustible fuels. The good news is that mankind has found a way to use digital technologies to optimise and improve the usage of renewable energy, by providing the necessary visibility of consumption of fuels to daily users and help them plan better.

How modern technologies complement renewables

Energy management has been an important use case for technologies like the Internet of Things (IoT) and data analytics. So far, we are aware of smart plugs and sensorised devices that can analyse and notify us about the electricity being utilised inside buildings. Now, solution providers are coming up with devices that can monitor renewable energy directly.

Smart grids and net-metering are being taken up by discoms rapidly to make the overall energy transaction scenario transparent. Smart grids, along with smart meters, help discoms keep an eye on the generation and distribution statistics of energy in real time.

These also give them macro- and microscopic analytics into the distribution and consumption trends of energy